Feb 8, 2025

Image Denoising

- Noise Models
- MMSE filters (Wiener filters)

Types of Noise:

1. Gaussian Noise
2. Salt & pepper noise
3. Photon counting
4. Speckle
5. Quantization

⋮

Types of Models

x = I/P image

y = O/P image

z = Noise

Additive: $y = x + z$

Multiplicative: $y = xz$

x & z may / may not be statistically independent.

Gaussian Noise:

$$p(z) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(z-\mu)^2}{\sigma^2}\right]$$

Sum of large no independent random variables, converges to a Gaussian (By CLT)

Salt & Pepper Noise:

$$p(y=x) = 1-\alpha \qquad p(y=MAX) = \frac{\alpha}{2} \qquad p(y=MIN) = \frac{\alpha}{2}$$

Image Denoising for Gaussian Noise

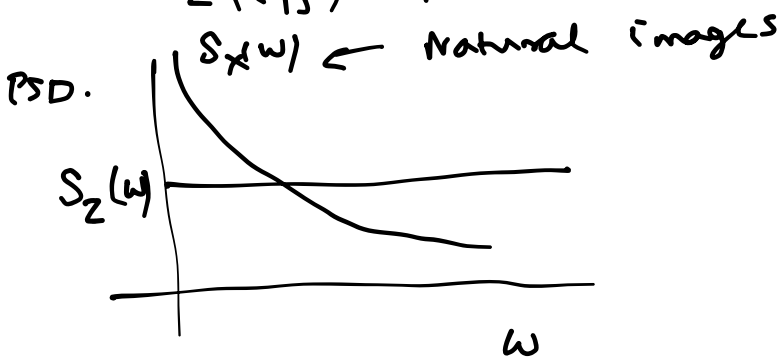
$$Y = X + Z$$

$$Y(i,j) = X(i,j) + Z(i,j)$$

O/P image I/P image Noise

$z \sim N(0, \sigma_z^2)$ $\qquad$ z is independent of signal.

$z(i,j)$ is i.i.d.



Low pass coefficients have higher signal to noise ratio when compared to high pass coefficients

$\Rightarrow$ We try to use low-pass filters for denoising.

$$y(i,j) = x(i,j) + z(i,j)$$

$$\mu_y(i,j) = \sum_{l=-P}^{P} \sum_{k=-P}^{P} w(k,l)\, y(i-k, j-l)$$

$(LPF(y) = \mu_y)$

$w \equiv$ filter coefficients of low pass filter.

$LPF(x) \equiv \mu_x \qquad LPF(z) = \mu_z$

$$\mu_y(i,j) = \mu_x(i,j) + \mu_z(i,j)$$

$$\hat{x}(i,j) = \mu_y(i,j)$$

$\mu_y(i,j) \approx \mu_x(i,j)$
high SNR

<u>Low pass filters:</u>

1) Rectangular windows.

$$h(i,j) = \begin{cases} \dfrac{1}{(2P+1)^2} & -P \le i,j \le P \\ 0 & \text{e.w.} \end{cases}$$

$$H(\omega,\lambda) = \underline{\sin[(2P+1)\omega]} \quad \sin. \quad \ldots$$

1D : $W(\omega) = \sum_{-P}^{P} e^{-j\omega m} = \ldots$

2. Ideal Low-pass filter:

$$H(\omega,\lambda) = \begin{cases} 1 & |\omega| \le \omega_c \quad |\lambda| \le \lambda_c \\ 0 & \text{e.w.} \end{cases}$$

— $h(i,j) = \ldots$

3. Gaussian filters:

$$h(m,n) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{1}{2}\frac{(m^2+n^2)}{\sigma^2}\right]$$

$$H(\omega,\lambda) \approx \exp\left[-\sigma^2(\omega^2+\lambda^2)\right]$$

<u>Wiener filter (MMSE).</u> (Minimum mean squared error).

Can we also recover the high frequency components?

$$Y = X + Z$$

$$\mu_Y = \mu_X + \mu_Z$$

$$Y - \mu_Y = X - \mu_X + Z - \mu_Z$$

High pass info in Y ....

$$\Rightarrow \quad Y_1 = X_1 + Z_1$$

$$Y_1 = Y - \mu_Y \qquad X_1 = X - \mu_X \qquad Z_1 = Z - \mu_Z$$

from Y_1, we would like to estimate X_T.

$$\hat{X}_1 = f(Y_1) \qquad\qquad X = \underset{\sim}{\mu_X} + \underset{\sim}{X_1}$$

$$\hat{X} = \mu_Y + \hat{X}_1 \qquad\qquad \hat{X} = \mu_Y + \hat{X}_1$$

we now focus on $Y_1 = X_1 + Z_1$

$$\hat{X}_1 = f(Y_1).$$

Minimum mean squared error estimation.

$$\hat{X}_1^* = \underset{\hat{X}_1 = f(Y_1)}{\arg\min} \; E\left[(X_1 - \hat{X}_1)^2\right]$$

$$E\left[(X_1 - \hat{X}_1)^2\right] = \iint (x_1 - \hat{x}_1)^2 \, g(x_1, y_1) \, dx_1 \, dy_1$$

$$= \int \underbrace{\left(\int (x_1 - \hat{x}_1)^2 \, g(x_1 | y_1) \, dx_1\right)}_{E[(X_1 - \hat{X}_1)^2] \text{ is minimized if above term is minimized.}} \underset{\geq 0}{g(y_1)} \, dy_1 .$$

$$\min_{\hat{x}_1 = f(y_1)} \int (x_1 - \hat{x}_1)^2 \, g(x_1 | y_1) \, dx_1$$

$$\underset{b}{\arg\min} \int (x - b)^2 \cdot f(x) \, dx = E[X].$$

$$\int 2(x-b)(-1) f(x) \, dx = 0$$

$$\Rightarrow \underbrace{\int x f(x) \, dx}_{E[X]} = \int b f(x) \, dx = b$$

$$\Rightarrow b = E[X]$$

$$\Rightarrow \hat{x}_1 = E[X_1 | Y_1 = y_1]$$

$$\hat{x}_1 = \int x_1 \, g(x_1|y_1) \, dx_1 = E[x_1 | Y_1 = y_1]$$

$$\hat{x}_1 = E[x_1 | Y_1] = f(Y_1).$$

$$Y = X + Z$$

$$Z \to N(0, \sigma_z^2)$$

$$Y_1 = \underline{\underline{X_1}} + \underline{\underline{Z_1}}$$

If $Z \sim N(0, \sigma_z^2)$, what is the distribution of Z_1?

$$Z_1 = Z - \mu_Z = Z(i,j) - \mu_Z(i,j)$$

$$= Z(i,j) - \sum_{l=-p}^{p} \sum_{k=-p}^{p} w(k,l) \, Z(i-k, j-l)$$

P.T. Z_1 is also Gaussian distributed.

$$Z_1 \sim N(0, \sigma_{Z_1}^2)$$

$$\sigma_{Z_1}^2 = \left[(1 - w(0,0))^2 + \sum_{\substack{k \neq 0 \, l \neq 0 \\ k = -p \text{ to } p \\ l = -p \text{ to } p}} (w(k,l))^2 \right] \sigma_z^2$$

For today, we assume $X_1 \sim N(0, \sigma_{X_1}^2)$

$E[X_1 | Y_1]$.

$$E[X_1 | Y_1 = y_1] = \int x_1 \, g(x_1 | y_1) \, dx_1 = \int x_1 \frac{g(x_1, y_1)}{g(y_1)} \, dx_1$$

(X_1, Y_1)

$$\underbrace{g(x_1, y_1)}_{\text{Bivariate Gaussian.}} \quad X_1 \sim N(0, \sigma_{X_1}^2) \qquad Y_1 = N(0, \sigma_{X_1}^2 + \sigma_{Z_1}^2)$$

$$g(x_1, y_1) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} [x_1 \; y_1] \, \Sigma^{-1} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \right]$$

$$\Sigma = \begin{bmatrix} \sigma_{x_1}^2 & \sigma_{x_1}^2 \\ \sigma_{x_1}^2 & \sigma_{y_1}^2 \end{bmatrix} \qquad \sigma_{y_1}^2 = \sigma_{x_1}^2 + \sigma_{z_1}^2$$

$$E[x_1 y_1] = E[x_1(x_1 + z_1)] = E[x_1^2] = \sigma_{x_1}^2$$

$$g(x_1|y_1) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{\dfrac{\sigma_{x_1}^2 \sigma_{z_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2}}} \exp\left[\frac{-1}{\dfrac{2\sigma_{x_1}^2 \sigma_{z_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2}} \left(x_1 - \frac{\sigma_{x_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2} y_1 \right)^2 \right]$$

$$g(x_1|y_1) \sim \mathcal{N}\left(\frac{\sigma_{x_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2} y_1 , \frac{\sigma_{x_1}^2 \sigma_{z_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2} \right)$$

$$E[x_1|y_1] = \frac{\sigma_{x_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2} y_1$$ linear estimate.

$$\hat{x}_1 = f(y_1) = \frac{\sigma_{x_1}^2}{\sigma_{x_1}^2 + \sigma_{z_1}^2} y_1$$